Muhammad Ahsan Ali

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Experience

Sabre Corporation — **Senior Software Engineer**
Aug, 2023 - Present
Worked on the Sass product.
Perform micro-services, data scheduling, and many more.
Perform multiple integrations with different types of platforms.
Implemented a payment gateway that helps the user make their payment through gift cards
Designed scalable backend architecture supporting enterprise multi-tenant eCommerce systems.
Optimized large orders by improving processing performance and reducing serialization overhead.
Handle the security of the payment gateway by using public-private key asymmetric cryptography.
AWS Account Migration from one account to another AWS Account using Terraform and automation scripts.

Softaims — **Full Stack (ROR) Developer**
July, 2021 - July,2022
Developed the project's interactive User Interface using HTML, CSS, Bootstrap, UiKit, and Tailwind CSS.
Implemented a queue system that can handle up to 10m daily jobs using Redis and Sidekiq
Managed the code using the View-Model approach, which drastically simplifies the view part of the project
Implemented unit testing using Capybara gem, Factory bot gem, and Rspec Gem
Implemented task management module using Rails Hotwire Turbo.

Technical Skills

⇒ *Programming Languages*	Skilled and experienced in Full Stack JavaScript, Python, Ruby, and C/C++.
⇒ *Databases*	Skilled in designing and implementing databases in MySQL, PostgreSQL, and MongoDB
⇒ *Frameworks*	Experienced in Ruby on Rails, Node, Stimulus.js, Hotwire-Turbo, Bootstrap, Tailwind, UIKIT, Angular, React
⇒ *Version Control*	Well-versed in various version control systems like Git.
⇒ *Integration:*	Stripe, Twilio, SendGrid, Mailchimp, Sidekiq, Elasticsearch, Redis
⇒ *Agent:*	Google ADK, A2A Protocol, MCP Protocol, VertexAI, OpenAI APIs, OpenClaw
⇒ *AWS:*	EC2, Lambda, S3, Secret Manager, SNS, SQS, SES, Api Gateway, Load Balancer, Target Group, Vpc, SecurityGroups, RDS, Auto Scaling etc
⇒ *GCP:*	Compute Engine, Cloud Functions, Cloud Storage, Secret Manager, Pub/Sub, SendGrid, API Gateway, Cloud Load Balancing, Backend Service, VPC, Firewall Rules, Cloud SQL, Instance Groups, CloudTask, CloudJob
⇒ *Other:*	CI/CD, Terraform, Packer, Github Action

Projects

Tarn Product — **FinTech Product**
Tarn Global Project
Tools: Python, Nodejs, ReactJS, Google Adk, Agent2Agent Protocol, and Model Context Protocol, GCP,
Built AI agents in Python to automate compliance verification and accelerate approval workflows.
Developed proxy services in Python and backend logic in Node.js to manage secure data routing and integration with external systems
Designed user interfaces in React.js for monitoring AI agents, visualizing verification results, and providing actionable insights
Implemented Application-to-Application (A2A) protocols enabling secure, automated interactions between internal and partner systems.
Deployed Message Control Protocol (MCP) to ensure reliable, compliant data exchange across platforms.
Leveraged Google Cloud Platform for deployment, monitoring, and orchestration of AI agents and proxy services.
Transformed manual compliance processes into automated workflows, reducing verification time and minimizing human error.
Enabled actionable insights from automated verification, improving compliance efficiency and decision-making.

Techsembly — **TS Saas Product**
Sabre Project
Tools: Ruby on Rails, PostgreSQL, Elasticsearch, Redis, and Sidekiq, Angular, and Elasticsearch
Managed the code using the View-Model approach, which drastically simplifies the view part of the project using Angular.
Rails Up-gradation from 5 → 6 → 7 framework modernization with zero critical production downtime
Angular Up-gradation from 9 → 19 with zero new bugs
Implemented Elasticsearch on the Rails service to achieve 8 times higher throughput.
Implemented the queue system that can handle up to 10m jobs using Redis and Sidekiq.
Perform 3rd-party integration to achieve good results.
Implemented double layer Authorization on the API for security.

Ts GiftCard — **Ts GiftCard**
Sabre Project
Tools: Ruby on Rails Server, AWS.
Implemented the api Server so that it can be used as a 3rd party service.
Perform multiple version handling of code.
Implemented unit testing using the Capybara gem and RSpec gem.
Implemented double layer Authorization on the API for security.

Payment Gateway — **Gift Cards Payment Gateway**
Sabre Project
Tools: Ruby on Rails, PostgreSQL, Angular, Public Private Key Asymmetric Cryptography, Heroku, and AWS.
Build a server that can handle the gift card payment.
Implemented the integration package for using gift card payment with stores.
Implement a public-private key asymmetric cryptography process to improve the security of the payment gateway.

Stads
Softaims Project

Sports advertisement marketplace
Tools: Ruby on Rails, PostgreSQL, Stimulus, SendGrid Redis and Sidekiq Heroku.
Managed the code using the View-Model approach which drastically simplifies views part of the project.
Implemented an efficient cache access mechanism on top of Rails to achieve 8 times higher throughput.
Managed the code using the View-Model approach
Designed an environment pipeline on Heroku and AWS, and implemented CircleCI for code testing
Implemented unit testing using Capybara gem and Rspec gem and test coverage up to 90

TeamRunner
Softaims

Event Experience Platform
Tools: Ruby on Rails, PostgreSQL, Heroku
Upgraded its rails version from 2.0 to 6.0 version
Designed an environment pipeline on Heroku and AWS
Implemented unit testing using the Capybara gem and the Factory Bot gem
Write a deployment script that automatically runs all the rake tasks, migrations, and other services once the deployment is done

Behind the Wall Detection
Sep 2020 - Apr 2021

Human Being Detection Behind any obstacle using Radio Waves
Tools: Python, OpenCV, Signals Machine learning and IOT.
Developed an IoT-based device that sends and receives the radio frequency using Raspberry Pi ways that pass through the obstacle and reflect once they hit the human being
Caught the reflected waves, converted them into a heat map, and then converted them into an image
Train these images using CNN algorithm

▷ Education

BCS (Hons)
2021, FAST NUCES.

Specialization in Computer Science
Courses included Databases, Programming, Algorithms, Operating Systems, Artificial Intelligence, Machine Learning, Digital Image Processing, Computing, and Distributed Systems with various Management and Mathematical electives.

▷ Achievements

Silver Medalist
2017-2021

FAST NUCES.

Microsoft Imagine cup
2019-2020

Microsoft Imagine cup
National Finalist